A Details of Evaluation Metrics

Reliability Given an editing fact (*s = United States, r = President of, o = Donald Trump, o* = Joe Biden*), it could be regarded edited effectively if the edited model f^* assigns a higher probability to the statement "*The President of the United States is Joe Biden*" than the original prediction (*Donald Trump*).

Generalization Edited models should be able to recall the updated knowledge when prompted within the editing scope. For example, the paraphrased prompts like "*Who currently holds the office of President of the United States?*" or "*Who is the current president of the US?*" can be used. The edited model f^* is considered to have generalized successfully if it can recall the editing memory, in this case, "*Joe Biden*".

Locality The edited model f^* should remain unchanged in response to prompts that are irrelevant or outside the scope of its editing. For example, the answer to the question "*Who is the President of France?*" should still correctly be "*Emmanuel Macron*".

B Task Prompts

The prompts for each task were illustrated in Table 2.

Reasoning: Q: {QUESTION} A: Let's think step by step. {HINT} Therefore, the answer (arabic numerals) is:
NLI: {SENTENCE1} entails the {SENTENCE2}. True or False? answer:
Open-domain QA: Refer to the passage below and answer the following question. Passage: {DOCUMENT} Question: {QUESTION}
Closed-domain QA: Please answer the given question based on the passage. The answer should be exact 'yes' or 'no'. passage: {PASSAGE} question: {QUESTION}. answer:
Dialogue: Q: {ARTICLE} Which choice is correct? Answer Choices: (A){OPTION0} (B){OPTION1} (C){OPTION2} (D){OPTION3} A: Among A through D, the answer is
Summarization: {DIALOGUE} TL;DR:
NER: Please identify Person Entity from the given text. Text: {SENTENCE} Entity: Please identify Location Entity from the given text. Text: {SENTENCE} Entity: Please identify Organization Entity from the given text. Text: {SENTENCE} Entity: Please identify Miscellaneous Entity from the given text. Text: {SENTENCE} Entity:
Sentiment analysis: For each snippet of text, label the sentiment of the text as positive or negative. The answer should be exact 'positive' or 'negative'. text: {TEXT} answer:

Table 2: The prompts to LLMs for evaluating their zero-shot performance on these general tasks.

C Editing Methods & Datasets

Paradigm	Editing Methods	Additional Training	Batch Edit	Editor Parameters
Meta-learn	MEND	Yes	Yes	$Model_{hyper}$ + *L*MLP*
Locate-then-edit	KN	No	No	*L*neuron*
	ROME	No	No	MLP_{proj}
	MEMIT	No	Yes	$L*MLP_{proj}$

Table 3: Comparisons between several popular model editing methods following Yao et al. (2023). **Additional Training** refers to whether the methods need training before conducting specific edits. **Batch Edit** refers to editing multiple target knowledge simultaneously. **Editor Parameters** refers to the parameters that need to be updated for editing. *L* denotes the number of layers to update. *MLP* is FFN and MLP_{proj} is the second linear layer in FFN. *neuron* denotes the key-value pair in FFN.

Four popular editing methods as compared in Table 3 were selected to measure their performance in improving factuality as well as their impairment in the general abilities of LLMs, including: **KN** (Dai et al., 2022)[2] first selected neurons that were associated with knowledge expression via gradient-based attributions, and then modified MLP layer at the rows corresponding to those neurons by adding scaled embedding vectors. **MEND** (Mitchell et al., 2022a)[3] learned a hypernetwork to produce weight updates by decomposing the fine-tuning gradients into rank-1 form. **ROME** (Meng et al., 2022)[4] first localized the factual knowledge at a specific layer in the transformer MLP modules, and then updated the knowledge by directly writing new key-value pairs in the MLP module. **MEMIT** (Meng et al., 2023)[5] expanded the capabilities of ROME by enabling the editing of large amounts of factual data through the updating of a sequence of MLP layers.

All experiments were conducted using the EasyEdit tool (Wang et al., 2024a), ensuring standardized and reproducible evaluation. All editing instances were randomly sampled from the editing dataset.

The popular model editing dataset Zero-Shot Relation Extraction (ZsRE) (Levy et al., 2017) used in previous work (Cao et al., 2021; Meng et al., 2022; Yao et al., 2023) was adopted in our experiments. ZsRE is a QA dataset using question rephrasings generated by back-translation as the equivalence neighborhood. Each input is a question about an entity, and plausible alternative edit labels are sampled from the top-ranked predictions of a BART-base model trained on ZsRE.

[2]https://github.com/EleutherAI/knowledge-neurons
[3]https://github.com/eric-mitchell/mend
[4]https://github.com/kmeng01/rome
[5]https://github.com/kmeng01/memit

D Extensive Evaluation Results

D.1 Results of Side Effects of Model Editing

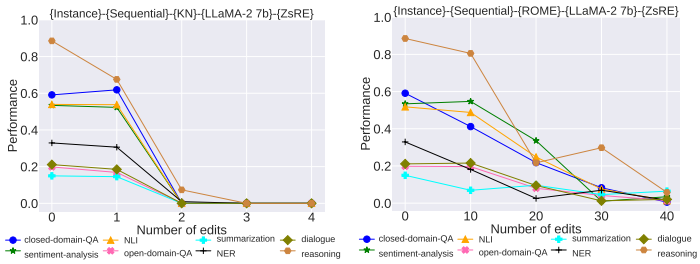

Figure 9: Performance on general tasks of edited models using KN or ROME to edit LLaMA-2 (7B) as the number of edits increases in *instance- and sequential-editing*.

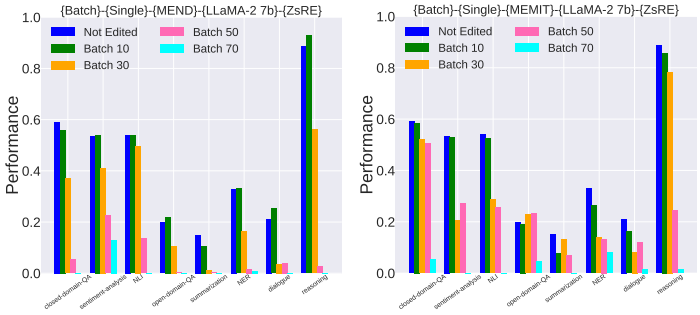

Figure 10: Performance on general tasks of edited models using MEND or MEMIT to edit LLaMA-2 (7B) with different batch sizes in *batch- and single-editing*.